

TrES-2b

R-1739

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190808 · created 2026-08-02T18:22:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458703.68292 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.04
Transit depth (Rp/Rs)^2	1.1 +/- 0.84 [%]
Orbital Inclination (inc)	87.83 +/- 2.94
Airmass coefficient 1 (a1)	1.414 +/- 0.024
Airmass coefficient 2 (a2)	-0.33 +/- 0.19
Scatter in the residuals of the lightcurve fit is	1.69 %
Best Comparison Star	#3 - [492, 186]
Optimal Aperture	3.34
Optimal Annulus	15.57
Transit Duration (day)	0.075 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.105 ± 0.04 vs 0.1254 (deviation 0.51σ)
Duration fit vs published	0.075 d vs 0.0746 d (deviation 0.03σ)
Mid-time O-C	-1.3 min
Depth SNR	2.6
Residual scatter	1.69 %

Light curve

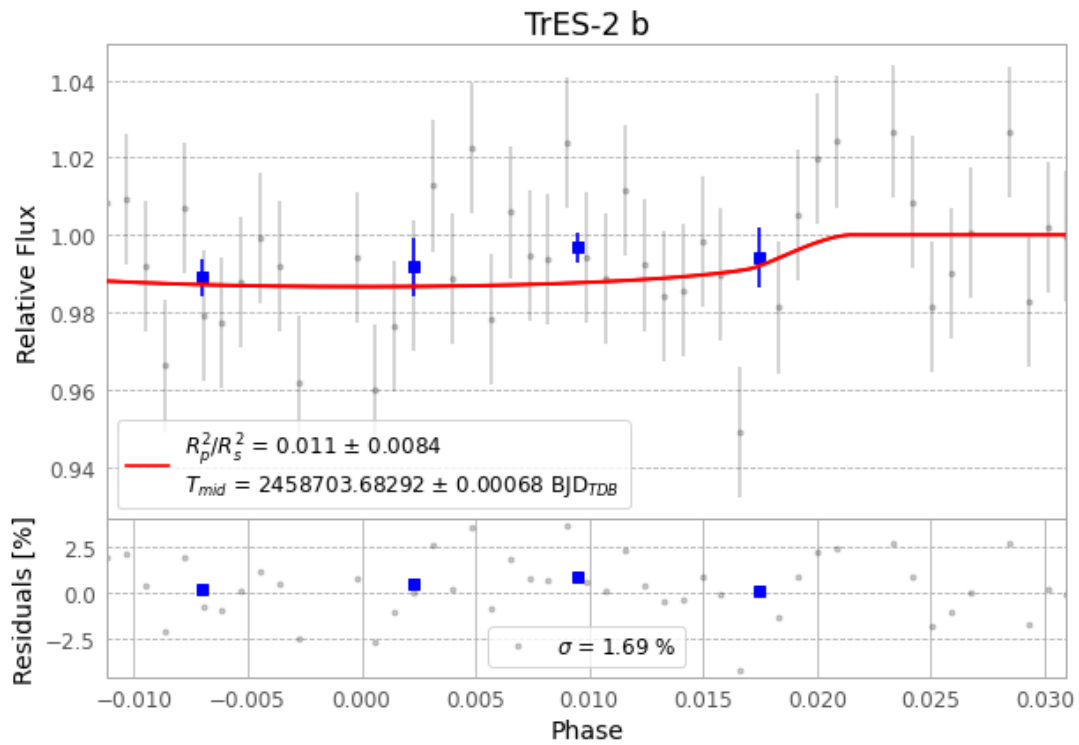


Figure 1 — FinalLightCurve_TrES-2 b_2019-08-07.png (SHA-256 27bf23d031053477...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [433, 25], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5718c35561d15d3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

49 FITS frames (32 MB), TRES-2190808033912.FITS → TRES-2190808060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190808.log (05d60ba53c9e...), FinalLightCurve_TrES-2 b_2019-08-07.png (27bf23d03105...), FinalLightCurve_TrES-2 b_2019-08-07.csv (b4fcce87ca88...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:22Z.